An automated method for generalised land cover mapping over New Zealand

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Summary

Project and client

The Ministry for the Environment (MfE, or 'the Ministry') sought to investigate opportunities to use new technology and data to find more cost-effective and timely ways to map land cover over New Zealand to meet the increasing needs for more timely information about where and how land cover is changing. The Ministry wanted to focus on automated mapping of a generalised set of land cover classes (n > 10), possibly to a finer scale (smaller minimum mapping unit) than the current Land Cover Database (LCDB v5).[1]

The Ministry contracted Manaaki Whenua – Landcare Research (MWLR) to develop a solution with the requirement that it be based upon open, regularly updated imagery, and as far as possible, on open-source code and systems.

Objectives

- Produce a basic land cover map as classified raster data, using existing, automated methods, from Sentinel-2 satellite imagery.
- Produce this basic land cover map using a generalised land cover taxonomy of around 15 classes selected because of their importance for land cover data requirements in New Zealand, and the feasibility of accurately identifying them with available data and established techniques.
- Aim for a high mapping accuracy of these land cover classes (i.e. an overall accuracy >95%).
- Map all distinct areas of land cover of at least 100 m^2 in size, and 20 m in width.
- Produce a map that covers the full terrestrial extent of mainland New Zealand and its offshore islands (includes Stewart Island / Rakiura, and Manawatāwhi / Three Kings Islands, but excludes the Chatham Islands and other outlying islands).
- Supply map data in the New Zealand Transverse Mercator (NZTM) projection at 10 m resolution.
- New mapping of wetlands, orchards and viticulture is out of scope; accept the results of LCDB for these classes by burning them into the map. These classes have a small areal extent and are relatively static.

Methods

- First, we implemented an automated method to generate a basic land cover map over New Zealand, consisting of 16 important land cover classes identified from publicly accessible Sentinel-2 satellite imagery. For this first step, we performed the three actions listed below.
  - Undertook accurate cloud-clearing and radiance standardisation of imagery enabling preprocessing of satellite imagery to seamless mosaics, enabling the application of spectral rules and deep learning to map land cover to high accuracy.

[1] See: https://lris.scinfo.org.nz/layer/104400-lcdb-v50-land-cover-database-version-50-mainland-new-zealand/

- Defined and applied spectral rules to delineate a hierarchy of land cover classes, to separate land from water, and then bare ground from vegetation. The vegetation was then split into herbaceous and woody vegetation.
- Applied deep learning to subtract exotic forest and deciduous hardwoods from the woody class, permitting the remaining to be divided into narrow-leaved scrub, broadleaved shrubs, and indigenous forest.
- Next, we undertook these additional steps.
  - Labelling of small patches of unclassified woody vegetation as unspecified woody.
  - Identifying croplands by the proportion of time they are bare over 5 years (2020-2024) of Sentinel-2 imagery.
  - Burning-in (i.e. accepting as true and not independently identifying) wetlands and vineyards from the LCDB since they are difficult to identify from spectral information.
  - Assessing overall accuracy by one expert validator examining a stratified random sample of points..
- Code and the trained models used to generate the map are available online at: https://github.com/manaakiwhenua/sentinel2-landcover-nz

Results

- We produced a 10 m resolution land cover map with 15 classes, which includes a sample colour palette for interpretation.
- This map's overall accuracy was 96%.
- Not all classes were accurate to the 95% threshold. However, almost all are more than 90% accurate. A full confusion matrix is available with precision and recall values for each class based on a stratified random sample of 4,500 locations.

Discussion

- We have shown that it is feasible to produce a generalised land-cover map (15 classes) for New Zealand, at 10 m resolution, rapidly, and with >95% overall accuracy.
- This study does not necessarily indicate how accurate a map using a more detailed set of classes would be.
- We had limited success training a deep learning model on small stands of trees at a small scale and scaling that up to a national data product.

Conclusions, applications

The high overall classification accuracy of the land cover map (96%) demonstrated here means that it could be useful for several important applications in New Zealand. These include the applications listed below. The first five points are uses for which we judge the map to be fully suitable; for the last two it is somewhat suitable.

1 Protective cover for soil erosion (woody class required, the spatial pattern is important as there is a need to combine information with slope derived from a digital elevation model).

2 National carbon accounting (gross area of land cover classes is an important metric).

3 Predator control (fine spatial detail of trees, i.e. corridors of suitable land cover for habitat patch connectivity).

4 Nutrient budgeting for water quality (delineation of cropping and orchards for modelling nutrient input to groundwater).

5 Habitat loss (monitoring loss of protected wetlands).

6 Fire fuel assessment (need for classes that indicate flammability, identifies the difference between shrubs and trees, the spatial pattern is important, and omission errors are worse than commission errors).

7 On-farm carbon accounting (gross area of land cover classes at farm scale is an important metric).

Recommendations

- We recommend the use of our draft land cover map over the use of global land cover data sets.
- We recommend that once a post-processing method has been established this final method is applied to Sentinel-2 mosaics to produce a national land cover map on an annual basis.
- We recommend that future work trial ideas (more details in main report) to remove small, isolated patches, address errors in identification of exotic forestry on remote offshore islands, distinguish between bare (built) and bare (natural), and perform change detection.
- We recommend checking for persistent land cover change over several years, checking for similar change in adjacent areas, or referring to a higher authority, the LCDB, where expert operators check all possible change.

Caveats to this work

- There is still a direct dependency on the LCDB as an existing, validated land cover map with a human-in-the-loop for two particularly difficult but critical classes: wetlands and viticulture.
- On a per-class basis, the 'broadleaved shrub' and 'unspecified woody vegetation' classes have considerably poorer validation results (~70% accuracy) than the others (>90%) and should be treated with more caution.

1 Introduction

National land cover mapping is a core component of New Zealand's environmental monitoring system. Since 1996, national land cover maps have been produced every five years and are collectively known as the Land Cover Database (LCDB).[2]

LCDB data are used in a wide range of applications including:

- environmental reporting indicators
- wildfire threat analysis
- land use modelling
- erosion risk modelling
- threatened environments analysis
- land use mapping for international greenhouse gas reporting (Land Use and Coverage Area frame Survey Analysis [LUCAS] programme).

Each update to the LCDB is time consuming and costly to produce since automated and semi-automated change detection is checked and verified before being accepted into the 5-yearly time steps. Though it has value as a heavily human-curated data set, its 1 ha low resolution (by modern standards) and return period can limit its application.

Many stakeholders have expressed the need for a timelier land cover mapping product. This could either be delivered through existing global land cover data sets[3] or through the development of ways to automatically map New Zealand's land cover from freely available satellite imagery.

Manaaki Whenua – Landcare Research (MWLR)[4] has a suite of existing in-house techniques for land cover classification from remotely sensed imagery, including necessary pre-processing. This report describes how these have been amalgamated to produce a basic land cover map over New Zealand using a cloud-free summer mosaic of publicly accessible Sentinel-2 imagery as the object of classification. The process is rapid (takes only several hours to complete), and repeatable.

The process consists of several distinct steps, which are described in more detail in Section 4.

1 Satellite image acquisition and pre-processing:

2 acquisition

3 reflectance normalisation

4 mosaicking and cloud detection

5 Defining and applying spectral rules for a binary split classification.

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[2] See: https://lris.scinfo.org.nz/layer/104400-lcdb-v50-land-cover-database-version-50-mainland-new-zealand/.

[3] This is considered in an associated report, see: Law et al. 2025.

[4] On 01 July 2025 Landcare Research New Zealand Ltd became the New Zealand Institute for Bioeconomy Science Ltd; Manaaki Whenua – Landcare Research operates as an internal group within this institute.

6 Training a deep learning classification model for the identification of exotic forestry.

7 Applying logical rules for combining intermediate data from prior stages.

8 Burn-in of LCDB classes (wetlands; orchards and vineyards).

Code and trained models used in the production of this layer is available on GitHub with an open licence at: https://github.com/manaakiwhenua/sentinel2-landcover-nz. However, the summer mosaic (processed Sentinel-2 imagery) is not publicly available.

2 Background

The Ministry asked MWLR to produce draft land cover maps for three regions of New Zealand (Auckland, Manawatū-Whanganui, and Canterbury). However, as the MWLR processes do not use locally-adaptative models or techniques (excepting separate North and South Island deep learning models), we developed a national data set.

Our final land cover taxonomy, consisting of 15 classes (labels) is shown in Table 1.

Table 1. Land cover taxonomy, indicating the method of identification for each class (assigned a class no [value] and label [name] in Table 1), and a suggested colour for visualisation

Value	Label	Method	Description	Colour (RGB, HTML notation)
0	Unclassified	NA	Exterior marine, or remnant from cloud masking and mosaicking.	#000000
1	Water	Spectral rules, binary split	Water bodies, including a coastal margin	#359ec9
2	Bare Ground	Spectral rules, binary split	Bare ground without vegetation cover	#d3d3d3
3	Indigenous Forest	Spectral rules, binary split	Mature forest comprising indigenous tree species	#006400
4	Herbaceous Vegetation	Spectral rules, binary split	Non-woody vegetation – primarily high- (exotic) and low-producing (exotic and indigenous) grasslands and indigenous tussock grasslands	#f6f99e
5	Cloud	NA (unused)	Cloud	#ff0000
6	Primarily Bare Ground	Spectral rules, binary split	Mix of bare ground (>50%) and vegetation cover	#d1b38c
7	Snow	Spectral rules, binary split	Snow cover at time of image acquisition	#ff00ff
8	Glacial Lakes, Wet Rock, Water/Sediment	Spectral rules, binary split	Glacial lakes, rock or sand covered by shallow water	#5ab2ff
9	Unspecified Woody Vegetation	Spectral rules, binary split	Small (<0.2 ha) clumps of unspecified trees	#666666

Value	Label	Method	Description	Colour (RGB, HTML notation)
10	Narrow-leaved Scrub	Spectral rules, binary split	Immature shrublands with narrow leaves – primarily mānuka, kānuka, gorse, broom, and matagouri	#709c63
11	Exotic Forest	Deep learning of Sentinel-2 imagery, trained on LCDB	Planted exotic forest – primarily Pinus Radiata (may include wilding pines)	#ab291f
12	Deciduous Hardwoods	Spectral rules, binary split	Mature exotic deciduous trees – primarily poplars and willows	#ffa600
13	Broadleaved Shrub	Spectral rules, binary split	Immature shrublands with broad leaves – primarily indigenous broadleaved species	#ff80a1
14	Croplands	Temporal sequence of NDVI (2020-2024)	Cropping land with evidence of regular tillage	#824ab3
15	Wetlands	Burned from LCDB	Wetland ecosystems	#91d491
16	Orchards and Vineyards	Burned from LCDB	Land managed to produce grapes and other perennial crops	#b44ab3

3 Objectives

- Produce a basic land cover map as classified raster data, using existing, automated methods, from Sentinel-2 satellite imagery.
- Produce this basic land cover map using a generalised land cover taxonomy of around 15 classes selected based on both their importance for land cover data requirements in New Zealand, and the feasibility of accurately identifying them with available data and established techniques
- Aim for a high mapping accuracy of land cover classes (user and producer accuracies) with an overall accuracy of >95% for each class.
- Map all distinct areas of land cover of at least 100 m^2 in size, and 20 m in width.
- Produce a map that covers the full terrestrial extent of mainland New Zealand and its offshore islands (includes Stewart Island / Rakiura, and Manawatāwhi / Three Kings Islands, but excludes the Chatham Islands and other outlying islands).
- Supply map data in the New Zealand Transverse Mercator (NZTM) projection at 10 m resolution.

4 Methods

We worked by adapting and applying established in-house methodologies to generate the basic land cover map over New Zealand from publicly accessible Sentinel-2 satellite imagery. Manaaki Whenua – Landcare Research (MWLR) has developed and applied these methodologies when mapping many of these classes in existing land cover products, using spectral rules (Dymond & Shepherd 2023) and deep learning encoder-decoder models (Martin et al. 2023).

4.1 Classification

4.1.1 Pre-processing

The image we used for the classification was a cloud-free, summer mosaic of Sentinel-2 satellite data for the summer of 2023/24 (i.e. 1 November 2023 to 29 February 2024 inclusive). This was pre-processed to have standardised spectral reflectance to ensure that this becomes a property of the vegetation alone, independent of sun position, slope, and view direction, as described in Dymond & Shepherd 2004.

We used the method imagery is described in Shepherd et al. 2020 for automated cloud-free mosaicking of Sentinel-2 imagery. This involves the use of an improved Tmask (i.e. multiTemporal mask) algorithm for cloud detection, and an object-based cloud and shadow approach to remove false cloud detections. For a given summer period in New Zealand, this uses around 100 contributing satellite overpasses and typically results in remaining cloud of less than 0.1%. Small areas of remaining cloud in the 2023/24 mosaic have been assigned class 0 (unclassified) in the output.

4.1.2 Spectral rules and binary splitting

Most land cover classes were identified by using a binary-splitting process with reference to expert-defined spectral rules. The conceptual method and reflectance values are described in Dymond & Shepherd 2004. Although that paper refers to Enhanced Thematic Mapper+ (ETM+) satellite imagery, the same concept applies to Sentinel-2 imagery. Table 4 in Dymond & Shepherd 2004 lists the spectral rules, with threshold values for mapping land cover from standardised spectral reflectance (denoted by ρ for a given band). We retained the values for ρ stated in that table, *except* for the split between narrow-leaved scrub and other woody vegetation. This had been defined by $(\rho_4 - \rho_5)/(\rho_4 + \rho_5)$ with a threshold value of 0.33; here we have instead used a threshold value of 0.25 (where ρ_4 is near-infrared [NIR, 0.77–0.90 μm]; and ρ_5 is short-wave infrared [SWIR, 1.55–1.75 μm]).

The selection of threshold values was determined interactively to achieve optimum spatial distribution based on expert judgement. More details about this process are given in section 4.1.7.

The application of a hierarchical set of binary split rules has several advantages, outlined below.

- Land covers are naturally organised into a hierarchy (e.g. vegetation vs non-vegetation).
- Different discrimination functions can be used at each decision node.
- Binary discriminant functions are simple to express and repeat.
- Land covers identified through other methods (e.g. the deep learning approach for exotic forestry) can be applied to results at suitable points in the hierarchical tree.

One example of the last advantage is to only accept the exotic forests identified using deep learning techniques for one branch of a binary discriminant function to limit its application. This approach was adopted in this study, as the binary classification rule for 'Forest' which was divided into 'exotic forest' (deep learning) and 'indigenous forest' (the remaining 'Forest'). This constraint prevented the exotic forestry identified with deep learning from over-ruling prior binary decisions. This ensures a more deterministic output and prevents the final identification of exotic forestry from being too similar to LCDB (i.e. the training data, which is produced at a relatively coarse 1:50,000 scale).

4.1.3 Temporal sequences (croplands)

We used a temporal sequence of normalised difference vegetation index (NDVI) computed using all available, cloud-masked, topographically normalised Sentinel-2 images over the January 2020 to December 2024 (inclusive) period to distinguish areas of cropland. The method used to produce this index is described in Amies et al. 2021 and involves the calculation of the median NDVI of vegetated pixels, median NDVI of all observations, and a ratio of time in vegetation and not in vegetation. When the temporal NDVI sequence is visualised as an RGB stretch (Figure 1), areas of cropland are distinct. One limitation of this technique is that it produces a signal likely to be associated with cropping (repeating periods of bare and vegetated land) but does not consider the time or times of the year at which this occurs. Thus, cropping is the probable explanation for the observed pattern, but not the only possible explanation in all cases. On the contrary, as it is based on a five-year temporal sequence, the identification is made over several years: it is not a single point-in-time identification of a process that is not strictly annual at field scale.

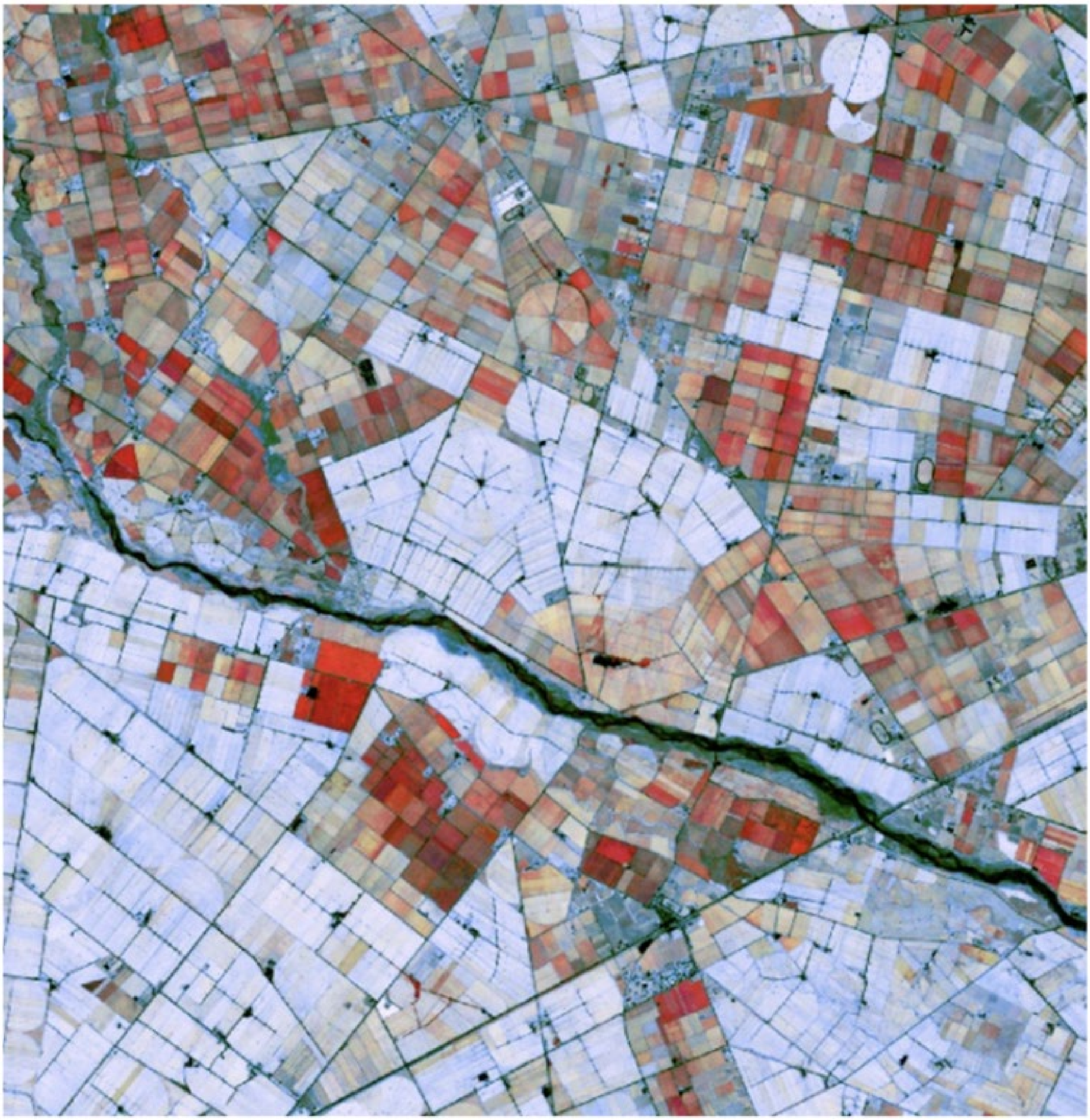

Figure 1. A map of Canterbury highlighting vegetation patterns indicative of rotational cropping (red) and permanent pasture (blue) using ratios of positive NDVI images. A red area suggests cropping as it indicates alternation between fallow and vegetated periods throughout the 5-year sequence.

4.1.4 Deep learning (exotic forest)

We used a U-Net (convolutional neural network for image segmentation) deep learning encoder-decoder network to train a model to generate a class raster for 13 classes (consisting of the initial set of classes proposed by MfE, plus a further separation of 'indigenous hardwoods' and 'deciduous hardwoods'). This further separation sought to explore whether deep learning can discriminate small patches of deciduous and evergreen trees. The network was trained using a class polygon layer derived from LCDB v5 using mapping suggested by MfE[5], except we also separated Broadleaved Indigenous Hardwoods, and Deciduous Hardwoods into their own classes to test whether they could be separated by the model. The suggested rules included the LCDB “Forest – Harvested” class being considered part of bare land and therefore excluded from the model training for exotic forest. Separate models were trained for the North and South Islands.

For imagery, various combinations of summer mosaics produced by MWLR were tried including: 2018/19; a stack of 2016/17, 2017/18 and 2018/19; 2023/24. The most accurate results when tested were obtained using models trained on the three-year image stack for the North Island, and 2018/19 only for the South Island, and these were used to generate the predicted land cover for 2023/24.

We trialled the model for different ‘widths’ (i.e. the number of filters in each layer), but this had negligible impact on performance while adding significantly to compute requirements. On balance, the original U-Net width of 64 filters in the initial layer was selected as a good compromise. The model was then trained for 1000 epochs[6] on image patches of size 512 × 512 pixels. These image patches were normalised to give a mean of zero and standard deviation of 1 over the entire mosaic. Our training used the Adam optimiser[7] to control the size of model updates, with an initial learning rate of 0.0001, and the model was evaluated against a validation set after every epoch. The model with the lowest validation loss was selected. The selected models were then trained on 50% of the imagery, validated on 10%, and tested on the remaining 40%.

There can be significant differences between mosaics depending on lighting and climate conditions for a given summer. We tested several approaches for maximising prediction robustness, which was visually assessed based on the quality of the 2023/24 predicted raster data; in particular, we attempted to minimise false positives of exotic forestry caused by differences in pixel distributions. This led to two different approaches for the two islands:

- North Island: train on images from three mosaics (2016/17, 2017/18 and 2018/19), normalising all image patches the same based on the average mean and standard deviation of all three mosaics.
- South Island: train on the 2018/19 mosaic only and normalise each individual image patch to (mean = 0, standard deviation = 1).

The difference between the two islands occurs because the mosaics for the South Island have larger differences caused by climate differences, whereas in the North Island it is image lighting differences that dominate.

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[5] Request for Proposal (RFP), Ministry for the Environment, Cost-effective land cover mapping methods, Ref. 1297-01-RFP, p9 Fig. 1

[6] An epoch is a unit of processing, where each epoch processes a set number of images in “minibatches” of a few images each, updating the model’s parameters after each minibatch. In this case, each epoch consisted of 300 minibatches, and two images were processed in each minibatch.

[7] The Adam optimiser (short for adaptive moment estimation) scales the effective learning rate in proportion to the ratio of the average error to the error variance. Early in training the average error dominates, and so updates are large. As training progresses, the variance begins to dominate, and the update size is reduced to prevent the model “overshooting” the optimal parameter set.

We assessed model performance by comparing the output to LCDB v.5.0 for both the 2018/19 and 2023/24 summer mosaics, the latter based on the assumption that the amount of change between land cover from LCDB v.5.0 to 2023/24 would be small, but also on the constraint of the present unavailability of LCDB v6.0.

Figure 2 shows that for the 2018/19 mosaic, the exotic forest class accuracy was 90.3% for the North Island and 83.5% for the South Island. Both models were able to separate exotic from indigenous forest with over 97% precision and recall (over 99% for the North Island). (The precision values reflect the conditional accuracy for these two classes given a pixel contains forest). The main class confusion for both exotic forest and indigenous forest was with the 'grass' class; however, there was also significant confusion with 'scrub,' 'bare' and 'indigenous hardwoods' classes.

North Island Recall 2018/19													
Truth/Predicted	**Bare**	**Snow**	**Water**	**Annual**	**Perennial**	**Grass**	**Wetland**	**Scrub**	**Ind. forest**	**Mangrove**	**Exotic forest**	**ec. hardwoods**	**Ind. hardwoods**
Bare	62.1%	0.0%	1.5%	0.0%	0.1%	15.4%	0.1%	1.7%	1.4%	0.0%	16.7%	0.2%	0.7%
Snow	92.2%	0.5%	7.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.1%
Water	2.7%	0.0%	89.8%	0.0%	0.0%	4.3%	0.4%	0.5%	0.9%	0.9%	0.1%	0.2%	0.1%
Annual	1.2%	0.0%	0.2%	16.4%	4.3%	77.1%	0.1%	0.0%	0.1%	0.0%	0.3%	0.1%	0.0%
Perennial	1.2%	0.0%	0.1%	2.4%	62.6%	31.6%	0.0%	0.1%	0.3%	0.0%	1.2%	0.2%	0.1%
Grass	0.5%	0.0%	0.1%	0.1%	0.2%	96.6%	0.1%	1.1%	0.6%	0.0%	0.5%	0.1%	0.2%
Wetland	1.6%	0.0%	3.3%	0.0%	0.2%	30.9%	40.9%	13.0%	1.3%	2.3%	3.2%	2.3%	1.1%
Scrub	0.5%	0.0%	0.2%	0.0%	0.0%	17.7%	0.5%	63.5%	10.5%	0.0%	1.9%	0.1%	4.9%
Indigenous forest	0.1%	0.0%	0.1%	0.0%	0.0%	2.9%	0.0%	2.6%	91.5%	0.0%	0.6%	0.0%	2.1%
Mangrove	0.5%	0.0%	13.3%	0.0%	0.0%	5.7%	6.0%	2.0%	0.7%	71.5%	0.2%	0.0%	0.0%
Exotic forest	2.8%	0.0%	0.1%	0.0%	0.1%	5.3%	0.0%	1.4%	1.4%	0.0%	87.6%	0.0%	1.2%
Deciduous hardwoods	2.1%	0.0%	1.6%	0.1%	0.6%	56.1%	3.1%	2.1%	3.7%	0.0%	4.2%	22.4%	4.1%
Indigenous hardwoods	0.7%	0.0%	0.4%	0.0%	0.1%	13.1%	0.1%	14.2%	32.4%	0.0%	3.6%	0.3%	35.2%
Accuracy	**87.0%**					**Recall | forest**			**99.4%**		**98.4%**		
						Precision | forest			**99.2%**		**98.7%**		

South Island Recall 2018/19												
Truth / Prediction	**Bare**	**Snow**	**Water**	**Annual**	**Perennial**	**Grass**	**Wetland**	**Scrub**	**Ind. forest**	**Exotic forest**	**Dec. hardwoods**	**Ind. hardwoods**
Bare	75.1%	0.5%	0.8%	0.0%	0.0%	19.8%	0.1%	1.5%	1.4%	0.6%	0.2%	0.0%
Snow	32.1%	67.5%	0.2%	0.0%	0.0%	0.2%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%
Water	8.9%	0.0%	85.7%	0.0%	0.0%	2.8%	0.5%	0.3%	1.5%	0.1%	0.2%	0.0%
Annual	0.2%	0.0%	0.1%	25.5%	1.1%	72.4%	0.1%	0.0%	0.0%	0.3%	0.1%	0.0%
Perennial	0.4%	0.0%	0.1%	2.1%	82.6%	14.0%	0.1%	0.1%	0.0%	0.3%	0.3%	0.0%
Grass	2.1%	0.0%	0.1%	0.4%	0.2%	93.6%	0.1%	2.5%	0.7%	0.2%	0.0%	0.0%
Wetland	1.2%	0.0%	3.4%	0.0%	0.0%	35.1%	47.6%	5.8%	6.2%	0.5%	0.2%	0.0%
Scrub	2.1%	0.0%	0.2%	0.0%	0.0%	32.9%	1.2%	50.0%	11.5%	1.6%	0.4%	0.0%
Indigenous forest	0.2%	0.0%	0.1%	0.0%	0.0%	1.4%	0.1%	1.9%	96.0%	0.3%	0.0%	0.0%
Exotic forest	1.8%	0.0%	0.1%	0.1%	0.1%	8.4%	0.1%	2.2%	3.0%	84.1%	0.2%	0.0%
Deciduous hardwoods	6.6%	0.0%	1.4%	0.2%	0.4%	40.3%	0.8%	7.9%	2.0%	3.6%	36.6%	0.0%
Indigenous hardwoods	2.4%	0.0%	0.6%	0.0%	0.1%	16.5%	0.3%	31.0%	42.4%	6.1%	0.6%	0.0%
Accuracy	**86.1%**					**Recall | forest**			**99.7%**	**96.6%**		
						Precision | forest			**97.0%**	**99.7%**		

Figure 2. Deep learning model performance (North and South Island models considered separately) in classifying n = 12 (South Island) and n = 13 (North Island, including 'mangrove' which does not occur in the South Island) land cover classes. Green shading indicates a correct classification. Orange shading is used to indicate misclassifications in the two forest classes above 1%. Rows contain the actual class, with columns representing the predicted classes (e.g. bare ground was predicted as snow <1% of the time). Green and red rectangles highlight the model's performance in separating indigenous and exotic forest: green rectangles highlight correct forest classification, while the red rectangles indicate an incorrect forest classification (given the pixel represents either indigenous or exotic forest).

4.1.5 Deep learning (specific woody classes)

Early in the development cycle, additional thematically detailed classes were considered (exotic conifer vs exotic non-conifer forests) but these were folded into a broader class due to lack of confidence in the class assignment. An attempt at distinguishing conifer/non-conifer trees (as opposed to forests) was also attempted using a deep learning model trained on aerial imagery at farm scale (Spiekermann et al. 2023). This was judged to be unsuccessful when applied to one-off regional aerial imagery at larger scale due to lack of ground truth information, highly variable image quality across the regions, unknown image survey dates and confusion between tree species of different age groups. Thus, such woody trees remain as "unspecified woody vegetation" in the final draft land cover map.

4.1.6 LCDB burn-in (wetlands, orchards, and vineyards)

Wetlands (LCDB class 15) and orchards and vineyards (LCDB class 16) were taken as given from the LCDB as these were considered too difficult to delineate automatically with sufficient accuracy.

4.1.7 Data integration following combinatorial rules

The ultimate sequence of thresholding rules, conditional application of exotic forest identification, and acceptance of LCDB (for wetlands, viticulture, and orchards) is available to read as code at our related GitHub repository.[8]

The repeatable workflow identifies land covers using binary decisions as described earlier, primarily from standardised reflectance mosaic imagery but the process also relies on ancillary data to make certain decisions at particular moments in the workflow.

The identification of snow, which is the first binary split, uses as ancillary data a 10 m national digital elevation model (DEM). This is used to eliminate any terrain under 150 m in elevation to avoid confusion between snow and bright coastal water (e.g. whitecaps and turbid estuary water). Subsequently, water is identified by considering the near-infrared (NIR), and short-wave infrared (SWIR/SWIR2) bands alongside the visual bands. The LCDB is used to conditionally alter the thresholds applied to these NIR, SWIR and red bands in the identification of woody areas (relaxing the thresholds wherever LCDB has pre-identified forests, mangroves, and scrub).

A centre/edge ratio computation is applied to identify small clusters of woody vegetation, so they are identified as "unspecified" woody vegetation due to insufficient pixel-level data to further classify these stands at 10 m resolution. Subsequently, the remaining woody vegetation is labelled by the deep learning model, limiting deciduous hardwoods to lowland areas, and applying NIR/SWIR threshold rules to distinguish narrowleaf and broadleaf forests from the woody layer.

Exotic forestry identified by the deep learning model (section 4.1.4) were included in the final land use map as disambiguation of forest identified in earlier binary splits. As the deep learning models were trained on 1:50,000 LCDB data, the models therefore learned to identify areas of bare land

[8] https://github.com/manaakiwhenua/sentinel2-landcover-nz

within forestry; yet this is a misidentification at this finer scale which we account for by only conditionally applying the results of the deep learning model.

Cropland (see section 4.1.3), the final binary split, is identified using a 5-year temporal median NDVI, conditional on land being identified as sufficiently flat to sensibly be cropland (i.e. based on the topography from the DEM). Wetlands, orchards, and vineyards are then accepted (burned in) from LCDB to form the final classified land cover dataset.

4.2 Validation

Validation was performed by taking a stratified random sample of pixels in the classified data. Pixels were selected at random, iteratively, and included in the sample if the sampled pixel was: i) in a non-excluded class for which there were still fewer than 300 samples; ii) if it had not already been sampled. This process continued until all classes were represented by 300 pixels. We used a constant random seed (the sample date) to be able to reproduce the sample, if necessary. The total sample size was 15 × 300 = 4,500 pixels, from a total population of 2.7 billion valid pixels. The spatial distribution of the sample is shown in Figure 3.

The pixels were all assessed by one expert validator, with experience in remote sensing and the New Zealand environmental context, who could label the pixels determinately as 'valid' or 'invalid,' or indeterminately as 'likely' or 'unlikely.' The indeterminate options allowed us to either exclude or include pixels that the operator found difficult to assess in the overall measurement of accuracy.

The validator used MapAccuracy software.[9] This presents sample points individually over the classified map, in parallel with a series of contextual windows set-up to display Sentinel-2 imagery (including the temporal sequence) with which to assess the classification. The operator could also consider recent aerial photography.

[9] See: https://anaconda.org/gillins/mapaccuracy

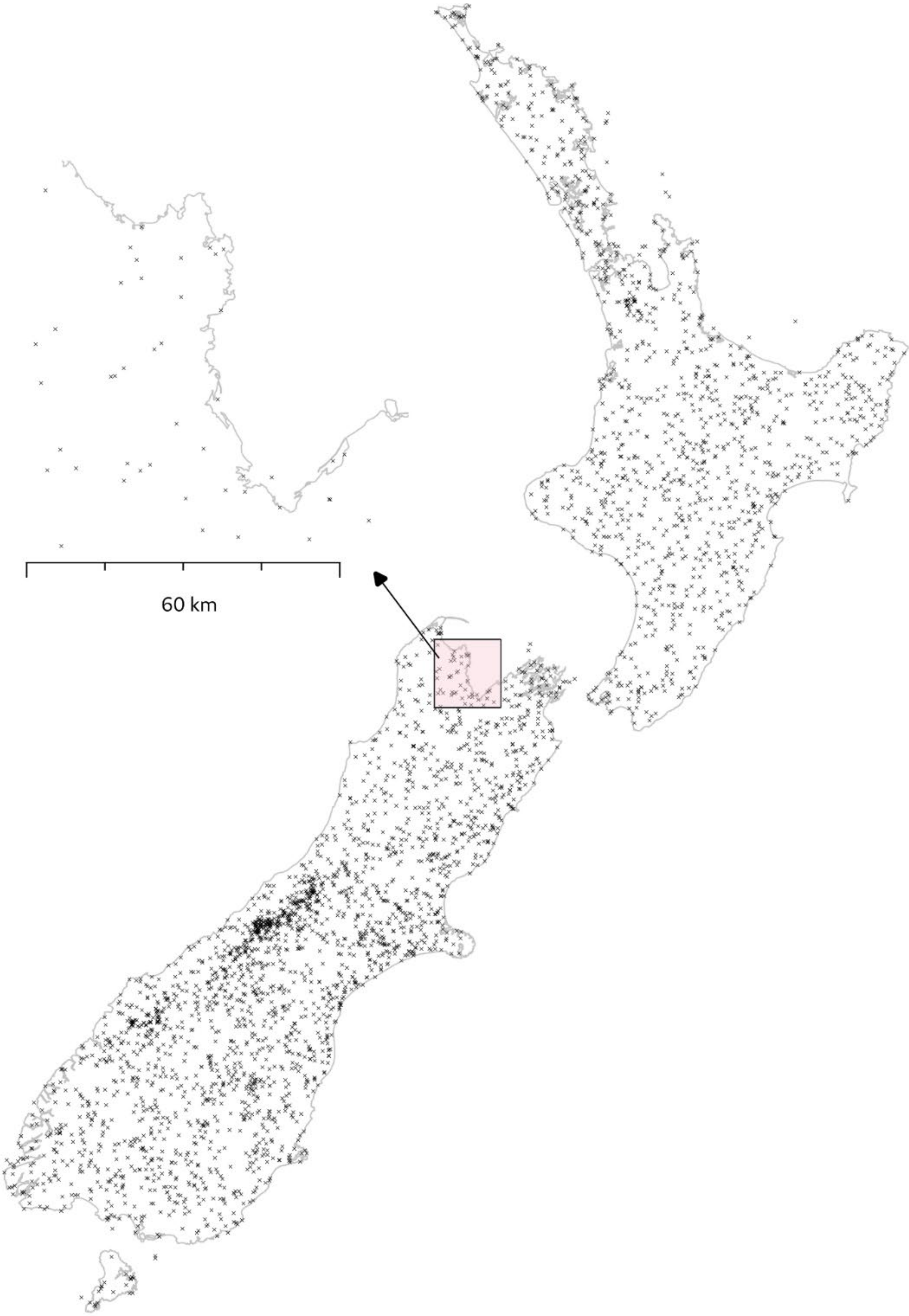

Figure 3. Map of the 4,500 randomly sampled (with stratification) points (black crosses), with inset map (pink) to show detail. Scale shown for inset map.

Some validation disagreement would be expected with different validators but, due to levels of resourcing, we were unable to perform validation with multiple redundant expert validators, or to use additional non-expert validators. (For a contextual example, Brown et al. 2022 found that consensus non-expert and consensus expert land cover validation for the Google Dynamic Earth land cover data set, which is also produced over Sentinel-2 imagery but has fewer classes, had an overall agreement of 77.8%.)

5 Results

The draft map with a suggested colour palette is shown in Figure 4. This is delivered in the ERDAS IMAGINE Hierarchical File Format (HFA), with a raster attribute table, in the NZTM projection.

Validation results are shown in Figure 5. The overall accuracy of this classification was 96.45%. All but two of the classes (unspecified woody vegetation, and broadleaved shrub) are >90% accurate. A confusion matrix is shown in Figure 6.

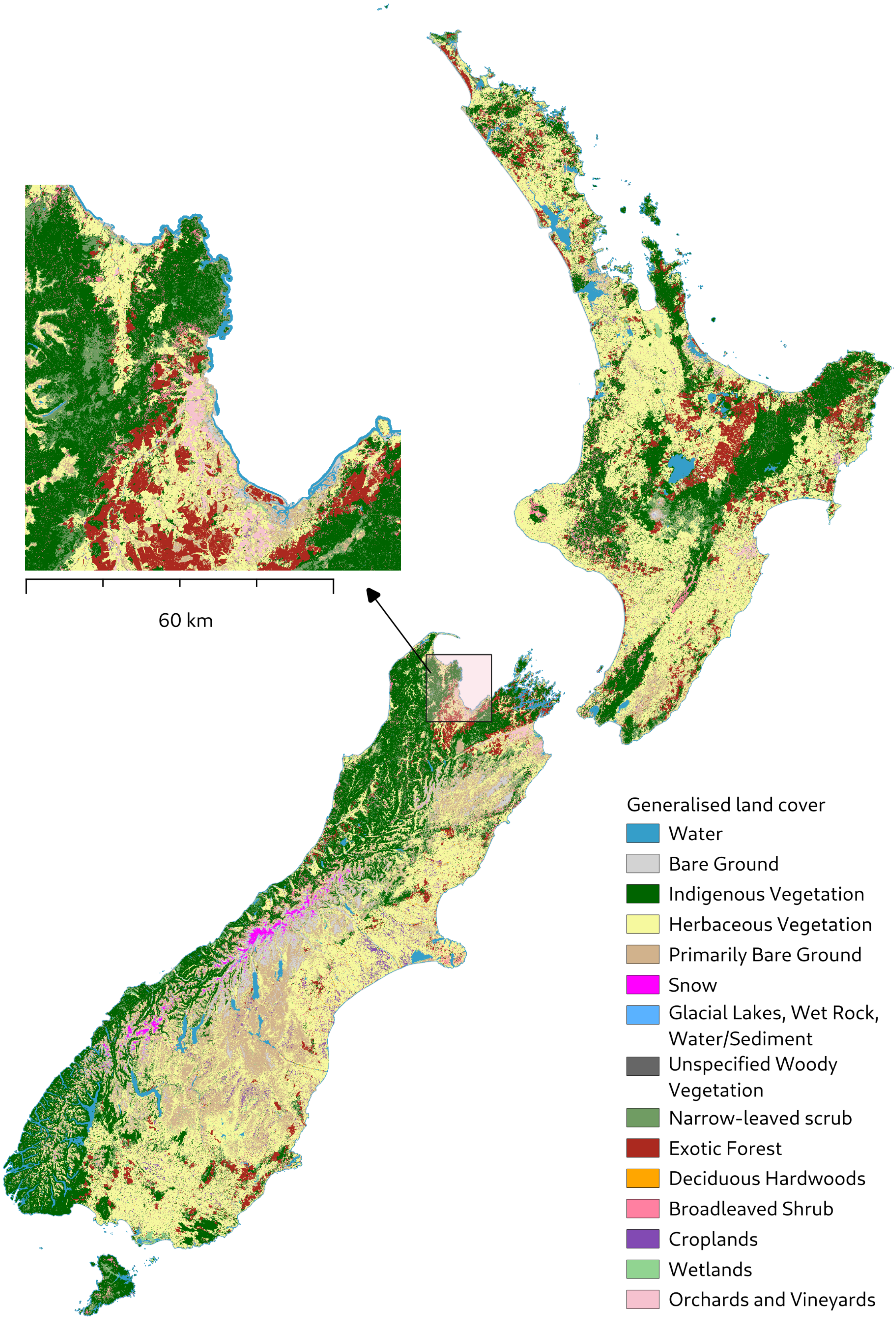

Figure 4.Draft land cover map (with inset map) produced as the final output of our process, showing the 15 land cover classes. Scale shown for inset map.

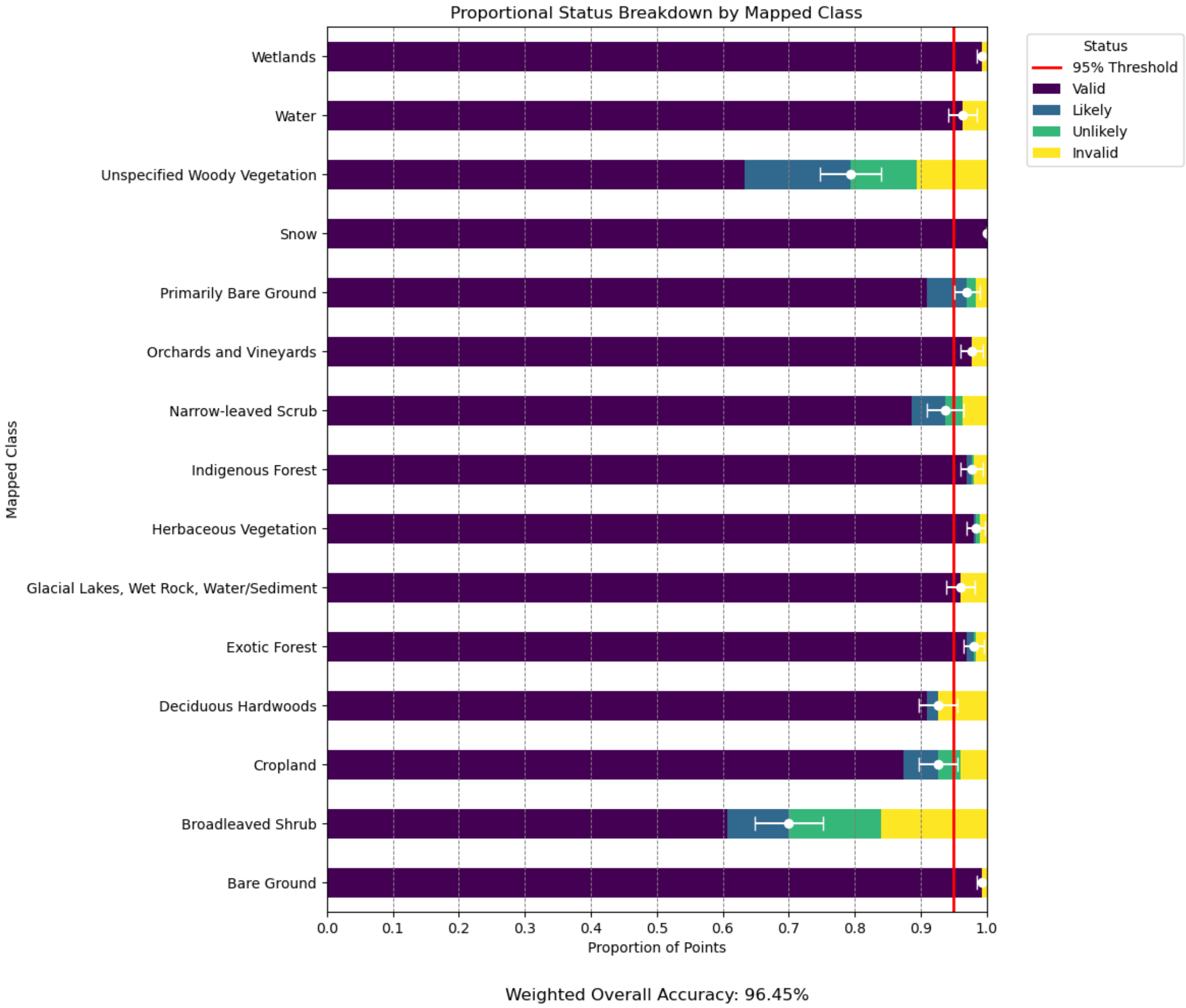

Figure 5. Validation results by class. The horizontal white error bar in each row shows the 95% margin of error. For the purposes of this figure, 'likely' results are assumed valid, and 'unlikely' results are assumed invalid, rather than excluding indeterminate results (though they are still visualised to give a sense of the limitations of remote validation). The consequent overall accuracy is 96.45%.

Predicted	Bare Ground	Broadleaved Shrub	Cropland	Deciduous Hardwoods	Exotic Forest	Glacial Lakes, Wet Rock, Water/Sediment	Herbaceous Vegetation	Indigenous Forest	Narrow-leaved Scrub	Orchards and Vineyards	Primarily Bare Ground	Snow	Unspecified Woody Vegetation	Water	Wetlands	Precision (%)	Recall (%)	F1-score (%)
Actual																		
Bare Ground	298	0	2	1	0	1	0	0	1	0	4	0	0	1	0	99.33	96.75	98.03
Broadleaved Shrub	0	210	0	1	0	0	0	0	0	0	0	0	0	0	0	70.00	99.53	82.19
Cropland	0	1	278	0	0	0	1	0	0	6	0	0	0	0	0	92.67	97.20	94.88
Deciduous Hardwoods	0	0	0	278	0	0	0	0	0	0	0	0	0	0	0	92.67	100.00	96.19
Exotic Forest	0	1	0	3	294	0	1	5	2	1	0	0	0	0	0	98.00	95.77	96.87
Glacial Lakes, Wet Rock, Water/ Sediment	0	0	0	0	0	288	0	0	0	0	0	0	0	0	0	96.00	100.00	97.96
Herbaceous Vegetation	0	26	17	9	1	0	295	1	4	0	1	0	40	0	2	98.33	74.49	84.77
Indigenous Forest	0	38	0	0	3	0	0	293	7	0	0	0	0	7	0	97.67	84.20	90.43
Narrow-leaved Scrub	0	11	2	7	2	0	0	1	281	0	0	0	0	0	0	93.67	92.43	93.05
Orchards and Vineyards	0	0	0	0	0	0	0	0	0	293	0	0	0	0	0	97.67	100.00	98.82
Primarily Bare Ground	0	8	1	0	0	0	1	0	2	0	291	0	9	1	0	97.00	92.97	94.94
Snow	0	0	0	0	0	11	0	0	0	0	0	300	0	2	0	100.00	95.85	97.88
Unspecified Woody Vegetation	0	0	0	0	0	0	2	0	0	0	0	0	238	0	0	79.33	99.17	88.15
Water	2	2	0	1	0	0	0	0	3	0	4	0	13	289	0	96.33	92.04	94.14
Wetlands	0	3	0	0	0	0	0	0	0	0	0	0	0	0	298	99.33	99.00	99.17

Weighted Overall Accuracy: 96.45%

Figure 6. Confusion matrix from a stratified random sample of 4,500 points (300 per class) over the final land cover dataset. To interpret precision and recall for each class, it is useful to think about an intended application. If false positives must be avoided, it is useful to value precision over recall. The inverse is true for false negatives. The F1 score is the harmonic mean of precision and recall, giving equal weight to each.

6 Discussion

Our method has produced a generalised land cover map (*n* = 15 classes) of New Zealand, at 10 m resolution, from a Sentinel-2 image mosaic, with processing times on the scale of hours and with >95% overall accuracy. However, note that the accuracy results do not necessarily indicate how accurate a map using a different, more detailed, set of classes could be.

We experimented with training a deep learning model on small stands of trees identified in aerial imagery on a small scale and then scaling that up to national-level data. We judged that this had limited success; as a result, isolated areas of trees remain labelled as 'Unspecified woody vegetation.'

7 Conclusions

The high accuracy (96%) of the cost-effective method for mapping land cover that we have demonstrated allows its use for five important use-cases (soil erosion, national carbon accounting, information for predator control, nutrient budgeting, and habitat loss); and its partial use for two use-cases (fire fuel, and farm carbon accounting.)

7.1 Applications

Some important New Zealand use-cases (i.e. applications) of land cover information are listed below.

1. Fire fuel assessment (classes that indicate flammability are required, such as the difference between shrubs and trees; the spatial pattern is important; for this application, omission errors are worse than commission errors)
2. Protective cover for soil erosion (a woody class is required; the spatial pattern is important because it is necessary to combine with slope from a digital elevation model [DEM])
3. National carbon accounting (the gross area in a region is an important metric)
4. On-farm carbon accounting (the gross area in farms is an important metric)
5. Predator control (the fine spatial detail of trees is important, i.e. corridors of suitable land cover for habitat patch connectivity)
6. Nutrient budgeting for water quality (delineation of cropping and orchards is important for modelling nutrient input to groundwater
7. Habitat loss (need to monitor loss of protected wetlands, and to have a useful wetland typology)
8. Soil carbon sampling (a useful distinction within bare ground would be between rocks and gravel [i.e. bare and no soil] and bare ground with soil).

7.2 Recommendations

Recommendation 1: We recommend the use of our draft land cover map over the use of global land cover data sets.

Table 2 compares the use-cases with those of the best global landcover data set (World Cover) according to Law and Dymond 2025.

Table 2. Comparison of the utility of NZ cost-effective land cover map, and the best available global land cover product (World Cover) for eight use-cases (applications) of land cover data in New Zealand. A tick (✓) means that the data is useful for the application, a cross (x) means not useful. A cross and tick means that the data is somewhat useful.

	World Cover data set	NZ cost-effective land cover map
Fire fuel	x ✓	x ✓
Soil erosion	x ✓	✓
National carbon	x ✓	✓
Farm carbon	x ✓	x ✓
Predator control	x	✓
Nutrient budgeting	x	✓
Habitat loss	x	✓
Soil carbon	x	x

Recommendation 2: We recommend that once a post-processing method has been established, this final method be applied to Sentinel-2 mosaics to produce a national land cover map on an annual basis.

This will be cost-effective because the method, from beginning to end, takes less than 10 hours of compute time to execute on available infrastructure (i.e. via New Zealand eScience Infrastructure [NeSI]).

Recommendation 3: We recommend that future work could trial ideas (listed below) to remove small, isolated patches, address errors in identification of exotic forestry on remote offshore islands, distinguish between bare (built) and bare (natural), and perform change detection.

1. To remove small, isolated patches.
    a. Majority filtering, to reduce noise by replacing the value of each pixel with the most frequently occurring class in its neighbourhood.
    b. Sieve analysis, to remove small, isolated patches of pixels that may be considered too small or insignificant, clumping similar classes together and removing noise.
2. To address errors in the identification of exotic forestry occurring on remote, offshore islands and various types of reserve land.
    a. Assign a different forest/woody class to exotic forests that are within such areas, as defined by auxiliary data.

3 To introduce a distinction between bare (built) and bare (natural).

   a Consider the definition of urban areas using auxiliary statistical boundary data, and assign accordingly, or;

   b Perform additional training for a deep learning algorithm for this purpose.

4 To perform change detection.

   a The first three post-processing corrections may assist in removing spurious change signals.

These, or similar, corrections could be applied on the change data set itself (as computed from un-post-processed data for different summer periods).

Although we considered various ideas for minor post-processing of the automated processing, we decided some of these might render the data less fit for some purposes whilst benefitting other purposes, so have not performed the processes; and additionally recommend that if they are performed, the less processed data should remain available for use.

Recommendation 4: We recommend checking for persistent change over several years, checking for similar change in adjacent areas, or referring to an authoritative source, the LCDB, where expert operators check all possible change.

While the land cover map has high spatial detail (10 m) and relatively high thematic resolution, making it useful for several important use-cases in New Zealand, care should be taken if cover change is inferred using simple differencing. Differencing will produce spurious change of small features even when maps are 99% accurate (c. 1% of NZ, which is a large area, will be inferred incorrectly as change each year). This effect should be considered seriously if differencing is necessary.

7.3 Caveats

There is still a direct dependency on the LCDB as an existing, validated land cover map with a human-in-the-loop for two particularly difficult but critical classes: wetlands and viticulture.

Broadleaved shrub and unspecified woody vegetation classes have poorer validation results than the others and should be treated with more caution.

8 Acknowledgments

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Declaration of the use of generative AI/AI-assisted technologies

This research work included the use of bespoke deep learning models as stated. Full details of these are given in the Methods section (Section 4).

Artificial intelligence was not used in the writing of this report.

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